

The operating system that runs between speech therapy sessions.

Wulo turns home practice into clinical evidence, so therapists open a session with data, not memory.

HOME-PRACTICE REVIEW

30 → 5 min

–83% reconstruction time at the start of a session

CLINICAL PHONEMES

+22%

Improvement over the off-the-shelf speech model on therapy targets

Wulo in one slide.

PROBLEM

30 minutes of a 60-minute session went to reconstruction.

Therapists had no reliable trace of what happened between appointments.

APPROACH

Automate only what is capturable, inspectable, and reversible by the therapist.

Diagnosis, goal setting, and discharge stay therapist-only. Everything else is drafted by the OS.

IMPLEMENTATION

Realtime voice + speech intelligence + admin layer.

Voice Live sessions, phoneme-aware assessment, AI session summaries, draft reports, parent emails, and SMS practice reminders.

RESULTS

25 minutes returned per session.

Pre-session reconstruction 30 → 5 min, plus a 22% relative reduction in phoneme error rate on clinical targets versus the off-the-shelf baseline.

A one-hour session was being eaten by missing context.

Therapists were forced to reconstruct the week before they could treat the child in front of them.

- Home practice was inconsistent, undocumented, and hard to verify.
- Parents wanted to help, but lacked time, structure, or clinical confidence.
- Children needed repeated practice without anyone available to coach it.
- The first 30 minutes of a 60-minute session disappeared into catch-up and guesswork.

THE HIDDEN COST

50%

of a one-hour appointment spent rebuilding the missing week.

Therapist-reported baseline, n = 3 SLTs, 12 sessions, Q1 2026.

I designed Wulo as an operating system for therapist-led care.

The question was not **what can AI do?** It was **what should the OS run on behalf of the therapist, and what must the therapist still own?**

01

Capture the signal.

If a practice event is not captured, it is invisible. I capture audio, transcripts, and pronunciation telemetry by default.

02

Compress for review.

Raw signal is not the deliverable. I run a pronunciation analyser that scores and summarises every session into a decision-ready record.

03

Therapist is the authority.

Every AI output is inspectable. Plans, scores, and notes are draft inputs; therapists approve, edit, or reject.

04

What I chose not to automate.

Diagnosis, goal setting, and discharge stay therapist-only. I never let the OS propose a clinical decision.

AUTOMATION CHOICES

What the OS runs for the therapist.

DECISION RULE Automate where the input signal is **capturable**, the output is **inspectable**, and the therapist remains the **authority**.

PRACTICE RUNTIME

Guided home sessions

Avatar-led voice practice gives a child structured repetitions without needing a parent to lead every turn.

CAPTURE LAYER

Transcript & audio

I stream, transcribe, and store every utterance, replacing memory and recall with a replayable record.

SPEECH INTELLIGENCE

Phoneme scoring

Per-word and per-phoneme pronunciation telemetry, tuned for clinical targets, not generic dictation.

REVIEW INTELLIGENCE

Session summaries

I compress effort, clarity, retries, and target accuracy into a clinical-grade review note.

PLANNING LAYER

Next-session drafts the therapist can approve.

I feed saved evidence and approved memory into a GPT-4o planner that proposes the next exercise plan for therapist sign-off. The therapist edits or rejects; nothing auto-applies.

NOT AUTOMATED

Diagnosis, goal setting, discharge.

These are clinical decisions, not workflow. I never draft them and never propose them.

The paperwork the OS quietly takes off the therapist's desk.

Clinical time was not only being lost **inside** sessions; it was being lost to the admin around them. Wulo treats reports, emails, reminders, and scheduling notes as first-class OS workloads, drafted from real session evidence and submitted for therapist approval.

REPORT WRITING · SHIPPED

Draft progress reports from real session data.

Transcripts, pronunciation scores, and AI session notes are compressed into a structured progress report. Therapist edits and signs.

EMAIL DRAFTING · SHIPPED

Parent and school updates, pre-written.

Wulo drafts the “how the week went” email per child, grounded in evidence, never invented. The therapist sends or revises in one click.

EXERCISE REMINDERS · PROTOTYPE

SMS nudges to families at the right cadence.

Targeted text reminders for the day's practice, driven by the therapist's plan, the child's history, and whether they actually practised yesterday.

ROUTINE ADMIN · DESIGNED

Scheduling, notes, follow-ups.

Session prep notes, post-session summaries, follow-up tasks, and waiting-list communications, generated, queued, and tracked by the OS.

SAME RULE, APPLIED TO ADMIN

Every draft is grounded in captured evidence, every output is inspectable, and the therapist still presses send.

What I chose, what I rejected, and why.

REALTIME STACK

I chose Azure Voice Live.

Rejected: OpenAI Realtime API (no avatar, weaker en-GB child speech), LiveKit + Whisper (own infra, no integrated TTS, higher first-token latency).

Why: I get avatar, realtime ASR, and TTS in one session contract, with one vendor to operate.

PRONUNCIATION PIPELINE

I chose Azure Speech PA with a custom PLS lexicon.

Rejected: Whisper + Montreal Forced Aligner (no phoneme confidence on child speech), bespoke acoustic model (cost and data unjustified at current scale).

Why: I get phoneme-level accuracy and reference IPA out of the box; PLS gives me the clinical /r/ target the default lexicon misses.

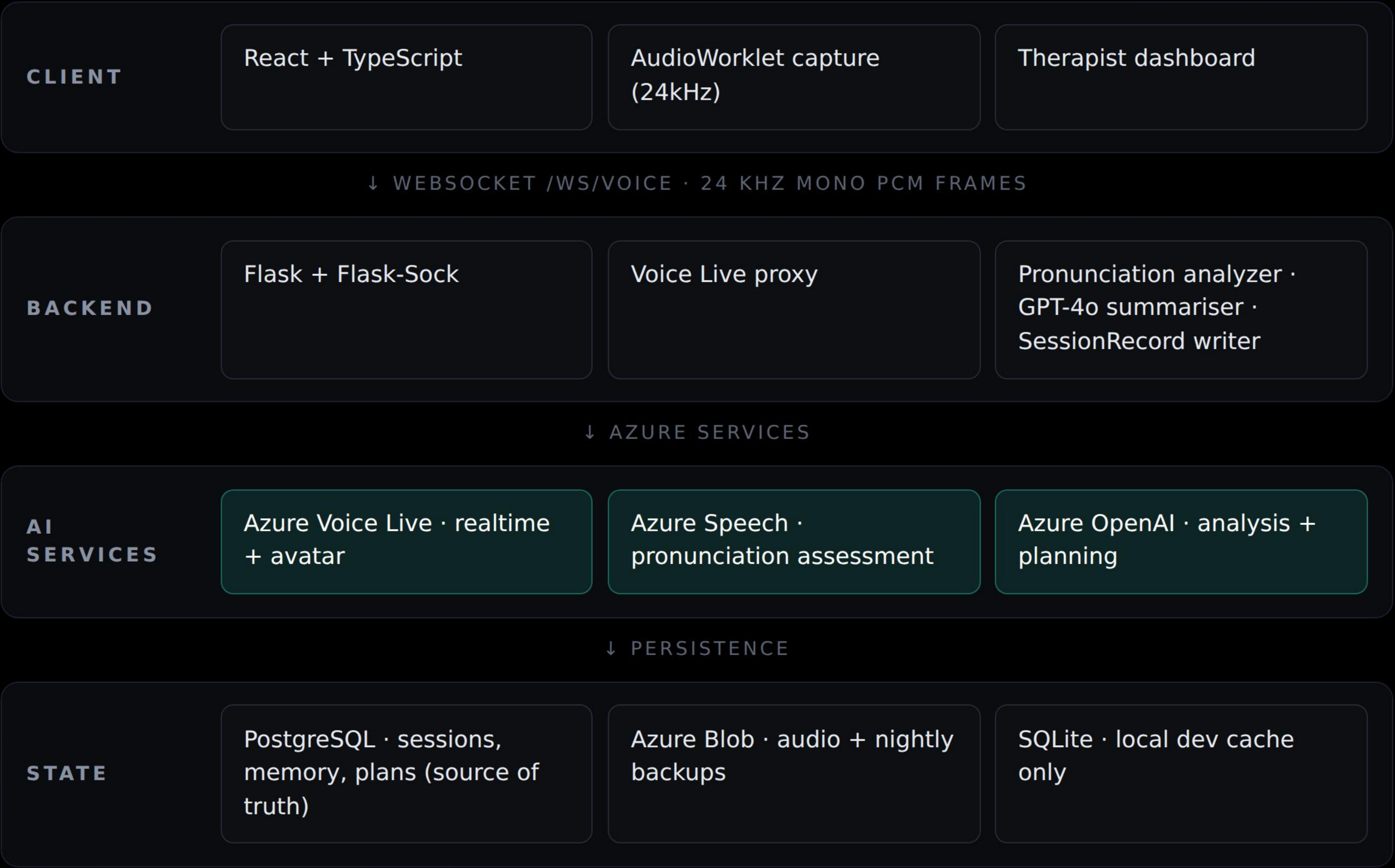
STORAGE

I chose PostgreSQL as source of truth.

Rejected: SQLite on Azure Files (SMB locking and corruption risk), Cosmos DB (overkill, weaker relational queries for session history).

Why: I get typed schemas, joins across child, session, and plan, and a managed HA path. I keep SQLite only as a local dev cache.

Two loops: realtime practice and asynchronous review.



REALTIME LOOP

I stream audio from the browser to a backend proxy. Voice Live returns assistant audio, avatar video, and transcriptions for the child to react to.

REVIEW LOOP

On session end, I score, summarise, and save the transcript, audio, and target words for the therapist to review before the next appointment.

The two records that pin the system down.

SessionRecord (Postgres)

```
session_id: UUID
child_id: UUID
started_at: timestamptz
duration_s: int
transcript: jsonb          # ordered turns
phoneme_scores: PhonemeScore[]
target_attempts: jsonb
audio_uri: text            # blob://...
ai_summary: text
therapist_notes: text
approved_at: timestamptz | null
```

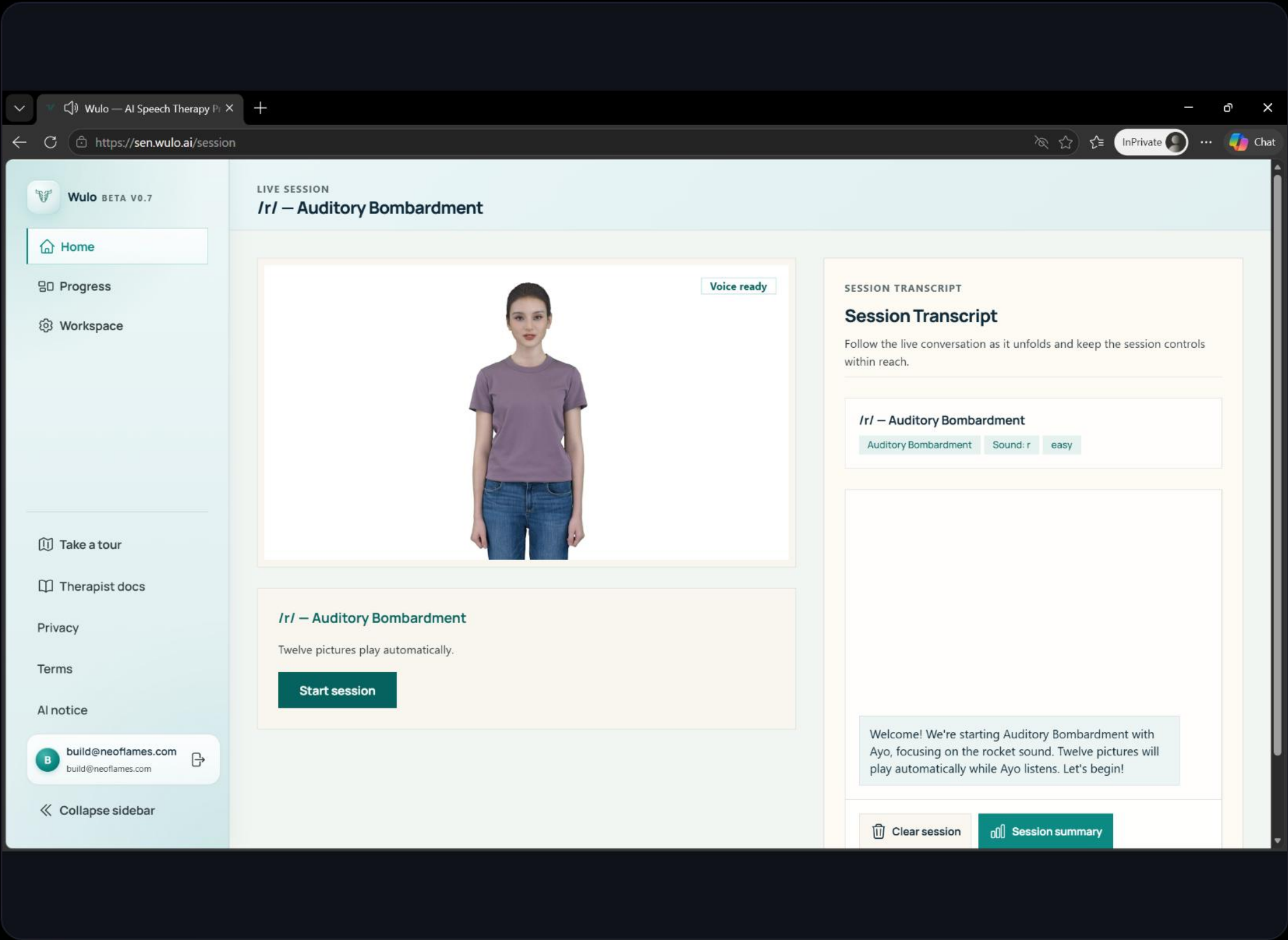
Written by the review loop on session end.

PhonemeScore

```
word: text
expected_ipa: text
observed_ipa: text
accuracy: float            # 0..1
error_type: enum(substitution, omission, distortion, addition, none)
```

Emitted per word by the Azure Speech analyser.

The child gets a practice partner. The therapist gets evidence back.



Live avatar-led speech practice session

LATENCY BUDGET

Target mouth-to-ear $\leq$ 850 ms p95: 20 ms capture frame, WebSocket hop $<$ 30 ms, Voice Live first token $<$ 350 ms, TTS playback start $<$ 450 ms.

- 1

Capture
AudioWorklet streams 24 kHz mono PCM in 20 ms frames to the backend over WebSocket. No WebRTC peer connection.
- 2

Converse
Voice Live returns assistant audio, avatar video, and per-turn transcription deltas. Server-side VAD with a 200 ms hangover lets the child barge in at any turn.
- 3

Score
Per utterance, I call Azure Speech pronunciation assessment to score accuracy, fluency, and completeness against target words, using a custom PLS lexicon.
- 4

Persist
I write transcript, phoneme scores, and AI assessment to Postgres as one typed SessionRecord; audio lands in Blob.

Making speech AI work for clinical phonemes.

Off-the-shelf speech models were not enough for the clinical /r/ target.
I built a phoneme-aware layer on top.

- Target-sound model linking exercises to specific words, phonemes, and acceptable substitutions for the child's age.
- Azure Speech pronunciation assessment for word-level accuracy, fluency, and completeness signals.
- SSML pronunciation lexicon (PLS) so /r/ targets are produced, and evaluated, with the right reference phonemes.
- I built an eval harness comparing baseline vs phoneme-aware behaviour on therapy-specific prompts.

RESULT ON THE /R/ TARGET

+22%

Relative reduction in phoneme error rate versus Azure Speech PA default lexicon, on the clinical /r/ target set.

Baseline: Azure Speech PA, en-GB, default lexicon.
Treatment: same model with custom PLS for /r/. Eval set: 160 utterances on /r/ targets, scored against an SLT-prepared reference (two raters, disagreements adjudicated).

What broke and what I learned.

HYPOTHESIS 1

Alias-only lexicon control.

Result: token synthesis collapsed into unrelated outputs such as variants of model and row.

Decision: rejected. Alias-only control was too weak to drive clinically sensitive drill speech.

HYPOTHESIS 2

Initial phoneme pass.

Result: directionally better than alias-only, but recognition was still unstable across the target blends.

Decision: not production-grade. Held back from the staging rollout.

HYPOTHESIS 3

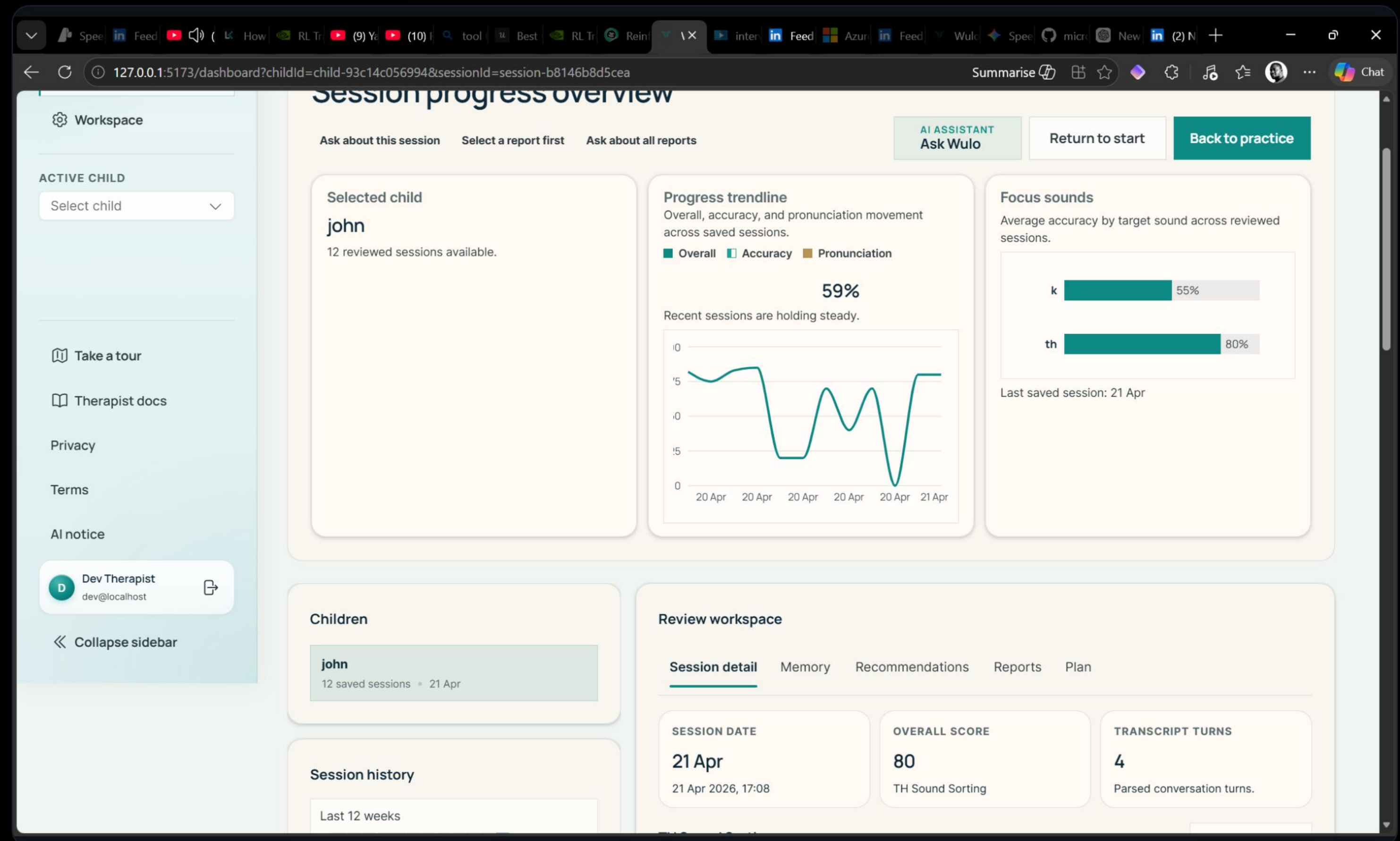
Best-candidate phoneme sweep.

Result: row strong, rah acceptable, roo and ree weak.

Decision: shipped as baseline; follow-up scoped to SSML drill lines only.

Raw practice becomes a decision-ready record.

- **SessionRecord:** session_id, child_id, started_at, duration_s, transcript[], phoneme_scores[], target_attempts[], audio_uri, ai_summary, therapist_notes.
- **PhonemeScore:** word, expected_ipa, observed_ipa, accuracy, error_type.
- **AI session note:** a draft summary of effort, clarity, retries, and a suggested focus for the next session. The therapist approves or edits before it is saved.
- **Therapist feedback** sits on top of the record; I never overwrite raw evidence.



Per-session pronunciation, AI assessment, and clinical notes

Measured impact.

BEFORE

30 min

Manual reconstruction at the start of a 60-minute session.

AFTER

5 min

Review using Wulo session evidence and AI summaries.

SPEECH QUALITY

+22%

Relative reduction in phoneme error rate on the clinical /r/ target, vs the off-the-shelf Azure Speech baseline.

METHODOLOGY

Reconstruction time: stopwatch-measured, 3 SLTs, 12 sessions baseline vs 12 sessions on Wulo, same caseload mix, Q1 2026.

Phoneme error: 160 utterances on /r/ targets, SLT-prepared reference, two raters with adjudication.

BOTTOM LINE Wulo returned **25 of every 60 minutes** of a therapy session to actual clinical work.

CLOSING

A speech therapy OS, not an AI therapist.

Wulo captures practice, analyses speech, summarises evidence, and proposes next steps. The therapist remains the decision maker, but now enters the session with better data and more time.

30 → 5 min

home-practice review

+22%

target phoneme performance

Next

per-child phoneme adaptation from
longitudinal SessionRecords